Harkunwar Singh

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HIGHLIGHTS OF QUALIFICATION

- Currently enrolled in **Level IV** of the **Honours Computer Science program** at **McMaster University**, expected to graduate in April 2026. Actively seeking **full-time new graduate** opportunities beginning May 2026.
- Proficient in **Python**, **Java**, **C/C++**, **Haskell**, **SQL**, **Elm**, **Assembly**, and LaTeX through hands-on projects and coursework.
- Skilled in database management and optimization using **MySQL** and **SQL**, enhancing operational efficiency and system automation.
- Strengthened **QA**, **DevOps**, and accessibility expertise as a **Co-op Student at the Ministry of Transportation (MTO)**, conducting **LIMS validation**, **streamlining CI/CD pipelines**, and improving portal usability.

EDUCATION

Honours Computer Science(Co-op) **September 2021 - April 2026**
McMaster University - Hamilton,ON

- **Engineering Award of Excellence:** Received the **McMaster Faculty of Engineering Scholarship ($3000)** for Academic Excellence.
- **Core Courses:** Data Structures & Algorithms, Software Development Life Cycle, Discrete Mathematics, Information Security, Operating Systems, Data Analytics, Computer Architecture, Software Design Using Web Programming, Statistics.

SKILLS

- **Programming Languages:** Python, Java, C/C++, JavaScript, TypeScript, SQL, Assembly, Haskell, Prolog
- **Frameworks, Libraries & Tools:** React, React Native, Expo, Node.js, FastAPI, Flask, Streamlit, NumPy, Pandas, Scikit-learn, MongoDB, Git, Linux, Docker, Jira, Confluence, Wireshark, PyCharm, Visual Studio, Figma, Microsoft Office
- **Cloud & Infrastructure:** AWS, Microsoft Azure, Netlify
- **Areas of Expertise:** Artificial Intelligence & Machine Learning, Natural Language Processing (NLP), API Integration, Data Structures & Algorithms, Encryption, Network Security

PROFESSIONAL EXPERIENCE

Engineering Associate Co-op **August 2024 - August 2025**
Ministry of Transportation Ontario - North York, Ontario

- Conducted **QA testing** and **validation** on the **Laboratory Information Management System (LIMS)** to ensure data integrity, system reliability, and regulatory compliance.
- Assisted with **DevOps** initiatives, optimizing CI/CD pipelines, and automating infrastructure to streamline deployments.
- Supported **lab automation** by identifying manual bottlenecks and integrating efficient, data-driven solutions.
- Collaborated with **QA, IT,** and engineering teams to strengthen data traceability, cross-functional workflows, and system efficiency.
- Wrote, maintained, and organized internal documentation and SOPs to improve knowledge sharing and operational consistency.

PROJECTS

NexDerm - AI-Powered Skin Lesion Detection App (Capstone Project) **September 2025 - Present**

- Built a cross-platform skin lesion detection app using **React Native, Expo, and TypeScript**, enabling users to upload images and receive real-time predictions.
- Developed a **FastAPI backend** with endpoints for image upload, preprocessing, inference, and structured prediction responses.
- Integrated and fine-tuned **EfficientNetV2/DenseNet CNN models** for skin lesion classification with confidence scoring and medical recommendations.
- Implemented reliable **frontend–backend communication**, robust error handling, and image storage pipeline, improving app stability and usability.

AI Resume-Job Matching System **January 2026 - Present**

- Designed and implemented an **explainable NLP-based resume–job matching system** using TF-IDF vectorization and cosine similarity to quantify compatibility between candidate resumes and job descriptions.
- Built a **custom text preprocessing and skill extraction pipeline** (tokenization, stop-word removal, n-grams) to identify overlapping and missing keywords while minimizing the impact of generic terms.
- Trained the TF-IDF model on a corpus of **real-world software engineering job descriptions**, improving domain relevance and reducing bias toward common, non-informative words.
- Developed a **lightweight scoring and analysis mechanism** that outputs match percentages, matched skills, and actionable resume improvement suggestions in a clear, interpretable format.

WeatherIQ **January 2025 - February 2025**

- Developed a modern weather web application using **HTML, CSS, and vanilla JavaScript**, delivering real-time weather insights through a clean, responsive, and visually engaging interface.
- Integrated the **Open-Meteo Geocoding and Forecast APIs** to display current conditions, 5-day forecasts, and hour-by-hour weather data with accurate location handling.
- Implemented **time-zone-aware date and time logic**, ensuring correct local time, sunrise/sunset times, and forecast day alignment for global locations.
- Designed interactive UI features, including **Celsius/Fahrenheit unit toggling, light/dark mode persistence, and an HTML Canvas-based temperature sparkline** to enhance usability and data visualization.